Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**AI-Driven Adaptive Process Optimization Platform for Smart Manufacturing**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (X), Next-gen Communication ( ), Advanced Robotics & Manufacturing (X), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	Robot Vision and Production Automation Technologies	R&D / Engineering Services	Testing / Inspection / Analysis Techniques
Concept of Proposed Technology	The proposed technology is a modular AI-driven process optimization platform designed to enhance efficiency, quality, and robustness in complex industrial processes. It integrates reinforcement learning for sequential decision-making, Bayesian optimization (see Figure 1) for parameter optimization, and AI-based expert systems for process monitoring and diagnostics. The technology has been applied to a variety of problems in manufacturing (see examples in Figure 2) It is currently adapted and extended to other industrial processes. The platform enables continuous learning and adaptation through digital twins and simulation-driven validation, making it suitable for deployment across manufacturing, robotics, semiconductor production, battery manufacturing, and in general energy-intensive industries. Given enough example data, this framework can execute the target process in an optimal way (see Figure 3).			
Rationale for Joint R&D Project	Korean SMEs face increasing pressure to improve productivity, yield, and energy efficiency while managing complex, high-dimensional processes. However, adoption of advanced AI optimization remains challenging due to limited in-house expertise and high experimentation costs. This joint R&D project combines the research institute's AI and simulation expertise with Korean SMEs' industrial know-how to co-develop validated, application-ready optimization solutions, reducing technical risk and accelerating industrial adoption.			
Key Words	1. AI Process Optimization 2. Reinforcement Learning 3. Bayesian Optimization			

	4. Digital Twins 5. Smart Manufacturing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 4	End of the Project	Level 7
R&D Objectives	• Develop a scalable AI optimization framework adaptable to the target industrial process • Integrate high-fidelity digital twins for safe AI training and validation • Achieve measurable improvements in productivity, quality, and energy efficiency • Demonstrate industrial feasibility through pilot implementations with Korean SMEs			
R&D Target Specifications	• Productivity improvement ≥ 15% • Quality defect reduction ≥ 20% • Reduction of experimental trials ≥ 30% • Real-time fault detection accuracy ≥ 95% • Robust adaptation to process drift without full retraining			
R&D Plan	Phase 1 – Process Definition and Data Preparation • Selection of target processes with Korean SME partners • Detailed process mapping and identification of control variables and constraints • Data acquisition planning, sensor validation, and baseline KPI definition Phase 2 – Digital Twin and Simulation Framework • Development of physics-based and data-driven digital twin models • Calibration and validation using historical and experimental data • Setup of scalable simulation environments for AI training Phase 3 – AI Optimization Model Development • Reinforcement learning for sequential decision-making and control • Bayesian optimization for parameter tuning • Safety constraints, reward design, and robustness testing Phase 4 – Pilot Deployment and Validation • Integration with existing industrial control and IT systems • Pilot-scale deployment and real-world performance evaluation • Model refinement addressing noise, drift, and operational variability Phase 5 – Industrial Readiness and Exploitation • Final performance validation and documentation • User interface and operational guideline development • Assessment of commercialization and scale-up potential			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Expected R&R of Research Institute: • Lead AI algorithm and system development • Design and validate digital twins and simulations • Ensure methodological robustness and safety • Support knowledge transfer and training
Expected R&R of Korean SME Partner	Expected R&R of Korean SME Partner: • Provide industrial use cases, data, and constraints • Support pilot implementation and validation • Provide feedback on industrial applicability • Participate in exploitation and dissemination

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will deliver validated AI-based optimization solutions enabling Korean SMEs to improve efficiency, reduce costs, and enhance competitiveness. It will accelerate digital transformation, support sustainable manufacturing practices, and establish a foundation for long-term international R&D collaboration and commercialization.

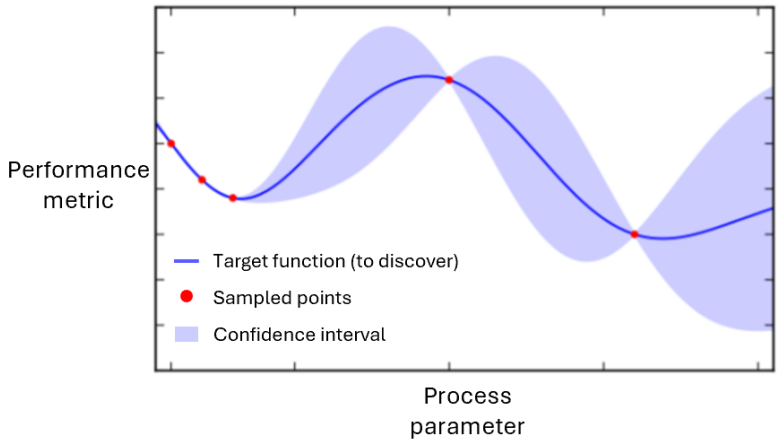

Figure 1: Illustration of gaussian processes applied on an input parameter to performance metric system

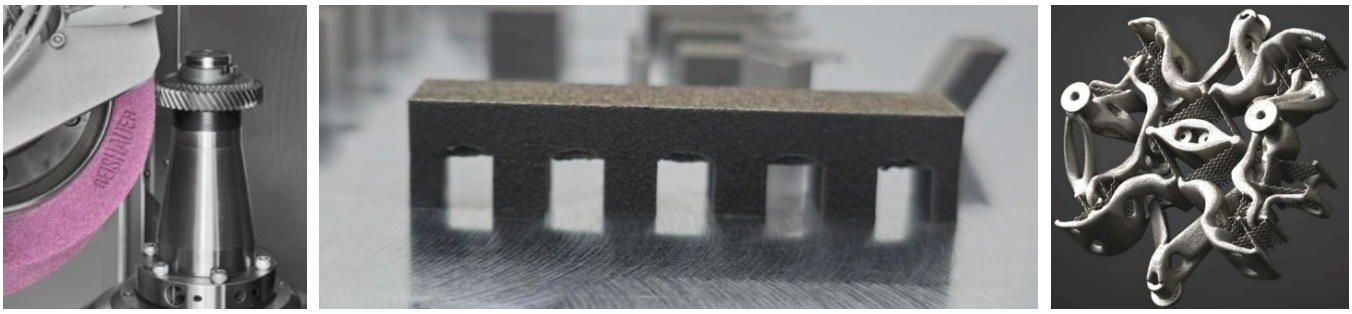

Figure 2: Examples of AI-process optimization application. From left to right: Continuous gear griding, Laser Powder Bed Fusion Additive Manufactruing, Compliant mechanisms for space applications

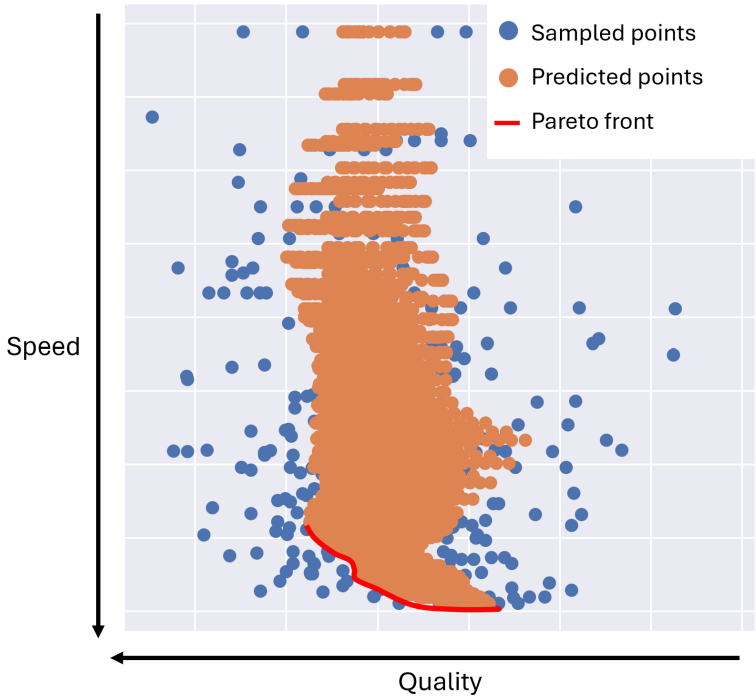

Figure 3: Identification of an optimal front after process prediction from smart sampling (from a successful project in the gear-grinding industry)

6. References

(1) GPyro: uncertainty-aware temperature predictions for additive manufacturing

I Sideris, F Crivelli, M Bambach - Journal of Intelligent Manufacturing, 2023

https://doi.org/10.1007/s10845-022-02019-7

(2) Too hot to print, too slow to handle; finding optimal path characteristics for WAAM

I Sideris, J Petrik, M Bambach - Manufacturing Letters, 2024

https://doi.org/10.1016/j.mfglet.2024.09.108

(3) CAM2S2 Compliant AM Mechanisms Systematic Synthesizer, Doctoral Thesis ETHZ, 2025

https://doi.org/10.3929/ethz-c-000791800

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization